

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			2 H ₂ O ₂ → O ₂ + 2 H ₂ O Right hand side (1) Balancing (provided products are correct) (1) Subscripts must be correct	1	1		2		
	(b)			Any 2 × (1) from: mass (of catalyst) (1) surface area (of catalyst) (1) concentration (of H ₂ O ₂)(1) volume (of H ₂ O ₂)(1)	2			2		2
	(c)	(i)		Drawing a reasonable tangent at t=30 (1) y step e.g. 10 and x step e.g. 15 (1) Approx. value of 0.5 - 0.7 (from candidate tangent)		3		3	3	3
		(ii)		Less reactants as reaction progresses (1) Less chance of successful collisions / less successful collisions per second (1) Accept: More dilute(1) Less particles so less successful collision (1)	2			2		
		(iii)		Start at origin and a steeper curve (1) Plateau sooner at 44 (1)		2		2		2
	(d)	(i)		Exothermic (1) because more energy is released when bonds are formed than required to {decompose hydrogen peroxide/break bonds} / Less energy in products than reactants (as energy released) / the energy released is greater than the activation energy (1)	2			2		
		(ii)		ΔH remains constant since products are the same (1) A catalyst lowers the activation energy (1) Which is E _a (1) so disagree Judgement must be given for 3rd mark			3	3		